OPD, Approach B, and TDM: Jacobians for trajectory distribution matching

Why swapping MSE for a score difference is not enough, and how truncated $\partial_\theta x$ recovers reverse-KL along an ODE grid

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Abstract

Companion to `dmd_opd_gradient_relation.tex`. We contrast four objectives used in XOPD-family distillation: (i) pointwise OPD field regression; (ii) endpoint DMD on a consistency sampler; (iii) *Approach B* — OPD's ODE grid with a score-difference force on trajectory states; (iv) Trajectory Distribution Matching (TDM). The critical invariants for (iii)–(iv) are an independent online fake score, differentiation through the sampler state ($\partial_\theta x_{t_i}$), and (for TDM) non-overlapping diffusion intervals. A residual swap that keeps OPD's fixed-x_t Jacobian is *not* reverse KL.

1 Jacobian table

Objective	Differentiates	Match object
OPD / `x0_norm`	$\partial_\theta v_\theta$ at fixed x_t	Teacher field (pointwise)
XDMD (endpoint)	$\partial_\theta x0_G$ (one consistency step)	$p_\theta(x_0)$ vs p_{real}
Approach B (`xopd_dm`)	$\partial_\theta x_{t_i}$ (one ODE step)	$p_\theta(x_{t_i})$ vs p_{real} on OPD grid
TDM (`xtdm`)	$\partial_\theta x_{t_i}$ (one ODE step)	Marginal KL on non-overlapping $[t_i, t_{i+1}]$

2 Why residual swap $\neq$ KL

OPD at fixed x_t minimizes $\mathbb{E}\,\|v_\theta(x_t) - v_\psi(x_t)\|^2$ (or the $\hat{x}_0$ form). Replacing the residual $v_\theta - v_\psi$ by $s_{\mathrm{fake}} - s_{\mathrm{real}}$ while still backpropagating only through the velocity *output* yields a hybrid with no clean reverse-KL identity: the score difference is a force in x-space, not a field target at a pinned input.

Genuine distribution matching (Diff-Instruct / DMD / TDM) uses

$$\nabla_\theta \mathrm{KL}(p_\theta \| p_{\mathrm{real}}) \approx \mathbb{E}\big[(s_{\mathrm{fake}}(x) - s_{\mathrm{real}}(x))\,\partial_\theta x\big],$$

with $s_{\mathrm{fake}} \approx \nabla \log p_\theta$ trained online. Pinning fake $\leftarrow$teacher zeros the signal; pinning fake $\leftarrow$student biases the score and collapses toward reweighted field matching through $\partial_\theta x$.

3 Approach B (ODE grid + score difference)

Keep the student as a multi-step flow on the OPD inference schedule $\{t_i\}$. At a randomly chosen index $i^\star$ (broadcast across ranks):

1. No-grad Euler rollout to $x_{t_{i^\star}}$.

2. One differentiable ODE step producing x (state carrying ∂_θ).
3. Diffuse $x_\tau = (1 - \sigma_\tau)x + \sigma_\tau \varepsilon$ with τ in the *non-overlapping* segment band $(\sigma(t_{i^\star+1}), \sigma(t_{i^\star})]$ (same local-τ rule as TDM; clamped to $[t_{\min}, t_{\max}]$).
4. Evaluate $\hat{x}_0^{\text{real}}, \hat{x}_0^{\text{fake}}$ under `no_grad`; form the DMD2 self-normalized direction grad $= (\hat{x}_0 - \hat{x}_0^{\text{real}}) - (\hat{x}_0 - \hat{x}_0^{\text{fake}})$ (normalized by mean $|\cdot|$).
5. Stop-grad identity / Pseudo-Huber on the *sample* so $\partial\mathcal{L}/\partial x =$ grad (up to Huber reweighting).

Memory is $O(1)$ transformer activations for the generator update (truncated BPTT). Relative to TDM, Approach B differs only by using the full OPD inference grid (`num_inference_steps`) instead of a fixed K-step grid (`tdm_sim_steps`).

4 TDM (Luo et al., ICCV 2025)

Paper objective (data-free, deterministic K-step generator):

$$\mathcal{L}(\theta) = \sum_{i=0}^{K-1} \sum_{\tau \in [t_i, t_{i+1}]} \lambda_\tau \, \mathrm{KL}\big(p_{\theta,\tau|t_i} \,\|\, p_{\phi,\tau}\big).$$

Intervals $[t_i, t_{i+1}]$ are *non-overlapping* so one fake network can separate trajectory segments by timestep. Practical surrogate: Pseudo-Huber on $\|x_{t_i} - \mathrm{sg}(\tilde{x}_{t_i})\|$ with $\tilde{x}_{t_i} = x_{t_i} + \lambda(s_{\text{real}} - s_{\text{fake}})$. Engineering (v1): match a *random* segment per generator step (same memory as Approach B); full sum over i is optional later.

5 Shared engineering invariants

- Independent online `fake` LoRA + manual data-parallel optimizer (outside ZeRO).
- Force acts on sample state x, never "MSE$(v_\theta, v_{\text{teacher}})$ with a score residual".
- Autocast weight cache off while swapping teacher weights; DDP bypass for `no_grad` weight swaps when applicable.
- Collective-safe random segment index (broadcast from rank 0).
- Eval uses the ODE sampler (multi-step flow), not XDMD's consistency schedule.

6 Relation to capacity

Under infinite capacity, OPD, Approach B, and TDM share the teacher field / induced trajectory as a common fixed point. Under a capacity gap, L^2 field projection and reverse-KL on trajectory marginals diverge: the latter is mode-seeking on the transported densities, which is why multi-time DM can outperform pure OPD for small students even when distilling the same deterministic teacher.